

Chapter 11

Mineral Resources

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

Minerals are natural substances having definite chemical and properties.

1. chemical
2. physical
3. biological
4. Both (a) and (b).

Answer

physical

Question 2

..... is a ferrous mineral.

1. Copper
2. Bauxite
3. Iron ore
4. Coal

Answer

Iron ore

Question 3

Good quality minerals are in quantity than low quality minerals.

1. equal
2. lesser

3. higher
4. poorer

Answer

lesser

Question 4

..... is one of the most widely distributed minerals found in the earth's crust.

1. Iron ore
2. Copper
3. Coal
4. Limestone

Answer

Iron ore

Question 5

Which ore of iron is reddish in colour?

1. Magnetite
2. Haematite
3. Limonite
4. All of the above.

Answer

Haematite

Question 6

..... is also known as 'Black ore'.

1. Limonite
2. Haematite
3. Magnetite
4. None of the above.

Answer

Magnetite

Question 7

..... is the best quality of iron that contains more than 70 per cent of iron.

1. Magnetite
2. Limonite
3. Haematite
4. Ironite

Answer

Magnetite

Question 8

About 90 per cent of iron ore is used to make

1. Bauxite
2. Steel
3. Brass
4. Bronze

Answer

Steel

Question 9

The main iron ore exporting ports in India are Mormugao and

1. Paradip
2. Haldia
3. Visakhapatnam
4. Kandla

Answer

Visakhapatnam

Question 10

..... is a raw material used for smelting of iron ore.

1. Manganese
2. Magnetite
3. Mica
4. Feronite

Answer

Manganese

Question 11

Manganese is used to make tough and resistant to rusting.

1. copper
2. bronze
3. bauxite
4. steel

Answer

steel

Question 12

..... is used in dry cell batteries.

1. Iron ore
2. Manganese
3. Copper
4. Bauxite

Answer

Manganese

Question 13

..... mineral is used to regulate blood sugar levels and to support the immune system.

1. Manganese
2. Iron ore
3. Mica
4. Copper

Answer

Manganese

Question 14

An excess of in plants and animals is toxic.

1. iron ore
2. magnetite
3. manganese
4. haematite

Answer

manganese

Question 15

..... is essential for plant growth and reduction of nitrates in green plants.

1. Mica
2. Iron
3. Manganese
4. Copper

Answer

Manganese

Question 16

..... is the earliest metal used by man.

1. Iron
2. Manganese
3. Bauxite
4. Copper

Answer

Copper

Question 17

Which of the following is a principal combination in which copper occurs in nature?

1. Sulphides
2. Oxides
3. Carbonate
4. All of the above.

Answer

All of the above

Question 18

..... is a good conductor of electricity, highly malleable and ductile.

1. Copper
2. Iron
3. Bauxite
4. Mica

Answer

Copper

Question 19

Copper is alloyed with and nickle to make stainless steel.

1. zinc
2. tin
3. iron
4. brass

Answer

iron

Question 20

Copper is alloyed with to make bronze.

1. iron
2. tin
3. nickle

4. zinc

Answer

tin

Answer the following questions

Question 1(a)

What is a mineral? State two characteristics of minerals.

Answer

Minerals are natural substances of organic or inorganic origin with definite chemical and physical properties.

Two characteristics of minerals are-

1. Minerals are not evenly distributed over space.
2. Minerals have inverse relationship in quality and quantity.

Question 1(b)

What is the difference between metallic and non-metallic minerals?

Answer

Metallic minerals	Non-metallic minerals
Metallic minerals are minerals in which metal elements are present in their raw form.	Non-metallic minerals do not contain any metal in them.
Metallic minerals are usually found in igneous and metamorphic rock formations.	Non-metallic minerals are often found embedded in mountains and sedimentary rocks.
Metallic minerals are good conductors of electricity as well as heat.	Non-metallic minerals are basically good insulators of electricity and heat.
Metallic minerals have high malleability and ductility.	Non-metallic minerals lack malleability and ductility. They break down easily.

Metallic minerals	Non-metallic minerals
Metallic minerals generally have lustre.	Non-metallic minerals do not have any shine or

Question 1(c)

Name the three varieties of iron ore mined in India. Name one state where each is found.

Answer

The three varieties of iron ore mined in India are-

- 1. Haematite** — It is found in Odisha.
- 2. Magnetite** — It is found in Tamil Nadu.
- 3. Limonite** — It is found in Uttarakhand.

Question 1(d)

- (i) Name the best quality of iron ore.
- (ii) Name two states where it is found.

Answer

- (i) Magnetite is the best quality of iron ore.
- (ii) It is found in Tamil Nadu and Karnataka.

Question 2(a)

What type of mineral is copper? Name any two states where copper is found in India.

Answer

Copper is a non-ferrous metal. It is found in Madhya Pradesh and Rajasthan.

Question 2(b)

State two uses of aluminium.

Answer

Aluminium is used in aircrafts and automobiles.

Question 2(c)

Give a reason for each of the following:

- (i) Manganese is an important raw material in the iron and steel industry.
- (ii) Bauxite is used in electrical equipment industry.
- (iii) Raw iron needs to be alloyed in order to be useful in construction and automobile industry.

Answer

- (i) Manganese is an important raw material in the iron and steel industry because it is used to make steel tough and resistant to rusting.
- (ii) Bauxite is used in electrical equipment industry because it is a good conductor of electricity.
- (iii) Raw iron needs to be alloyed in order to be useful in construction and automobile industry because raw iron by itself is not as strong and hard as needed for construction and automobile industry.

Question 2(d)

Bauxite is an oxide of which metal? Where are the deposits of Bauxite found in India?

Answer

Bauxite is an oxide of aluminium. The deposits of Bauxite are found in Goa, Odisha, Gujarat, Madhya Pradesh, Chattisgarh, Jharkhand, Maharashtra, Karnataka and Tamil Nadu.

Question 3(a)

How is brass and bronze made? State two uses of copper.

Answer

Copper is alloyed with zinc to form brass and with tin to make bronze.

Two uses of copper are-

1. It is used in building construction for roofing and plumbing.
2. It is used in ship-building industry.

Question 3(b)

Name two leading states producing manganese. State two industrial uses of manganese.

Answer

Two leading states producing manganese are Karnataka and Odisha.

Two industrial uses of manganese are-

1. It is an important raw material for the iron and steel industry.
2. It is used in chemical industries for manufacturing bleaching powder.

Question 3(c)

Give a reason for each of the following:

- (i) Magnetite is so called.
- (ii) Copper is used for making electric wires.
- (iii) Aluminium is used for making aircraft.

Answer

- (i) Magnetite is so called because it possesses magnetic property.
- (ii) Copper is used for making electric wires because it is a good conductor of electricity, highly malleable and ductile.
- (iii) Aluminium is used for making aircraft because it is light weight and strong.

Question 3(d)

- (i) Name two characteristics of iron ore deposits in India that facilitate their optimum use in the iron and steel plants.
- (ii) Name two industries that require iron ore as raw material.

Answer

- (i) Two characteristics of iron ore deposits in India that facilitate their optimum use in the iron and steel plants are-
 1. Iron ore deposits occur as hill masses and are easily accessible.
 2. They are found in close proximity to the areas producing coal, dolomite, limestone and manganese.
- (ii) Two industries that require iron ore as raw material are iron and steel industries.

Thinking Skills

Question 1

You have to prepare a feasibility report for establishing mineral based industries in your state? Which industries would you recommend? Give reasons to support your answer.

Answer

I would recommend developing iron and steel industry in my state because of the following reasons:

1. My state has adequate deposits of iron ore, coal, dolomite, limestone and manganese. This will cut down importing costs.
2. The area is well connected by roads and railways. Thus transportation will be hassle free.
3. Labour at reasonable wages can be employed from nearby areas.
4. Availability of a nearby port will facilitate export of finished products.

Question 2

Suppose you live in an area which has rich deposits of coal, dolomite, limestone and manganese and two exporting ports. Which industry can be developed in your area and why?

Answer

Iron and Steel industry can be developed in my area because of the following reasons:

1. The raw material will be easily available. Iron ore, fuel and flux supplies can be obtained at low costs.
2. The area has two exporting ports which means that it well connected with hassle free transportation.
3. Labour at reasonable wages can be employed from nearby areas.
4. The ports will make exporting of finished steel cost effective.

Question 3

Minerals are exhaustible over time. Make a list of household items that could be reused by repairing, recycling or refurbishing and the benefits of doing so.

Answer

The household items that could be reused by repairing, recycling or refurbishing are as follows:

1. Furniture — Old or damaged furniture can be repaired or refurbished instead of being thrown away.

2. Clothing — Clothes that are ripped or worn can often be repaired or repurposed into something new like cloth bags, pillow covers, bags or quilts etc. They can also be used as cleaning rags.
3. Cardboard boxes — They can be reused for storage or as a makeshift drawer divider.
4. Plastic containers — They can be reused for storing leftovers, packing lunch boxes, or as planters for small plants.
5. Electronics — Many electronics can be repaired instead of being replaced.
6. Kitchenware — Items such as pots, pans, and dishes can often be repaired instead of being thrown away.
7. Garden tools — Many garden tools can be repaired or refurbished instead of being thrown away.

The benefits of repairing, recycling or refurbishing household items are as follows:

1. Environmental benefits — Repairing, recycling, or refurbishing items reduces waste in landfills and conserves natural resources.
2. Cost savings — Repairing or refurbishing items can save money on buying new items.
3. Preservation of sentimental value — Repairing or refurbishing sentimental items such as furniture or toys can preserve their sentimental value and memories.
4. Skills development — Repairing items can develop valuable DIY skills and knowledge.
5. Contribution to the circular economy — By repairing, recycling, or refurbishing items, you are contributing to the circular economy, where resources are kept in use for as long as possible.